Total No. of Printed Pages:2

SUBJECT CODE NO: - RNP-8017
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A) (NEP) Sem - II
Examination November / December 2025 (To be held in February-2026)
Basics of Electrical and Electronics Engineering

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1. Use of non-programmable calculator is allowed
2. Q.no.1 is compulsory
3. Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 **Solve any 10 following questions** 20

1. Electric current is defined as
   a) Rate of change of voltage
   b) Rate of flow of charge
   c) Work done per unit charge
   d) Resistance per unit length

2. The SI unit of electric potential is
   a) Ampere b) Coulomb c) Volt d) Ohm

3. Ohm's law is valid for
   a) Non-linear conductors b) Semiconductors
   c) Electrolytes d) Metallic conductors at constant temperature

4. The unit of resistivity is
   a) Ω/m b) $\Omega \cdot m$ c) m/Ω d) $\Omega \cdot m^2$

5. Kirchhoff's Voltage Law is based on the law of
   a) Conservation of charge b) Conservation of energy
   c) Electromagnetic induction d) Coulomb's law

6. RMS value of AC current is equal to
   a) Peak value b) Average value
   c) $0.707 \times$peak value d) $1.414 \times$peak value

7. Power factor of a purely resistive circuit is
   a) 0 b) 0.5 c) 0.8 d) 1

8. Fleming's right-hand rule is used for
   a) Motors b) Generators c) Transformers d) Rectifiers

9. A Zener diode is mainly used as
   a) Rectifier b) Amplifier c) Voltage regulator d) Oscillator

10. Binary number system has base
    a) 2 b) 8 c) 10 d) 16

11. Which gate is known as Universal gate?
    a) AND b) OR c) NAND d) XOR

12. LVDT is used for measurement of
    a) Temperature b) Pressure c) Displacement d) Speed

13. The unit of magnetic flux density is
    a) Weber b) Tesla c) Ampere-turn d) Henry

SECTION B

Q2	a)	Explain Electric current, electric potential and potential difference with units.	05
	b)	State and explain Ohm's Law. Also derive expression for resistance in series and parallel.	05
Q3	a)	Explain Faraday's laws of electromagnetic induction and Lenz's law.	05
	b)	Compare AC and DC. Explain advantages of three-phase system over single-phase system.	05
Q4	a)	Explain PN junction diode: construction, symbol and V-I characteristics.	05
	b)	Describe BJT or JFET: types, construction and applications.	05
Q5	a)	Explain Basic blocks of a regulated power supply with neat block diagram.	05
	b)	Explain RTD, Thermocouple and Thermistor for temperature measurement.	05
Q6	a)	Explain series and parallel combination of resistances with diagrams.	05
	b)	State and explain Kirchhoff's Current Law and Voltage Law.	05
Q7	a)	Differentiate between active and passive components.	05
	b)	Explain electronic materials and types of semiconductors.	05